

Lecture 9: Functions and Countability

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1 Cartesian Products: Keeping Track of Order

Start with $A = \{1, 2, 3\}$ and $B = \{a, b, c\}$. An **ordered pair** (x, y) has a first coordinate x and a second coordinate y :

$$(x, y) = (u, v) \iff x = u \text{ and } y = v.$$

For sets A and B , their **Cartesian product** is

$$A \times B = \{(x, y) : x \in A \text{ and } y \in B\}.$$

In our example,

$$A \times B = \{(1, a), (1, b), (1, c), \\ (2, a), (2, b), (2, c), \\ (3, a), (3, b), (3, c)\}.$$

		B		
		a	b	c
A	1	$(1, a)$	$(1, b)$	$(1, c)$
	2	$(2, a)$	$(2, b)$	$(2, c)$
	3	$(3, a)$	$(3, b)$	$(3, c)$

Three choices for each coordinate give nine ordered pairs.

Two kinds of order. Order *within a tuple* matters: $(1, a) \neq (a, 1)$. However, the elements of a *set* are unordered. Writing $A = \{3, 1, 2\}$ gives the same set, and rearranging the nine pairs in the displayed list gives the same Cartesian product.

In $B \times A$, the first coordinate is a letter and the second is a number. In particular, $(1, a) \in A \times B$ but $(1, a) \notin B \times A$, so $A \times B \neq B \times A$. They have the same size, but different elements.

For finite sets, $|A \times B| = |A| |B|$: for each first coordinate, there are $|B|$ choices for the second. Here $|A \times B| = 3 \cdot 3 = 9$. The notation A^2 means $A \times A$, a set of ordered pairs. For example, $\{0, 1\}^2 = \{(0, 0), (0, 1), (1, 0), (1, 1)\}$. Also $A \times \emptyset = \emptyset$, since there is no second coordinate.

2 Functions: Exactly One Output for Each Input

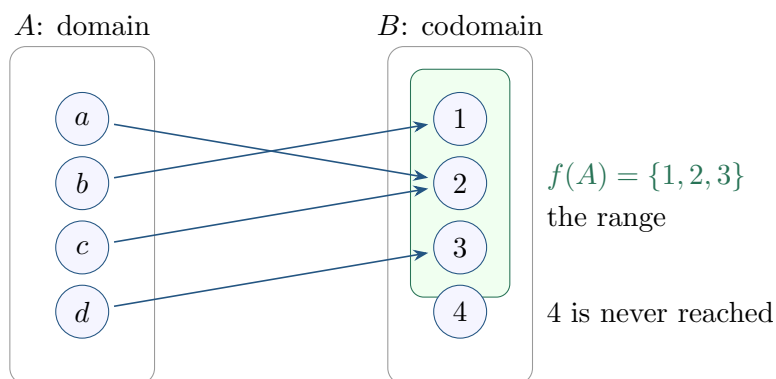
A **function** $f : A \rightarrow B$ assigns to every $x \in A$ exactly one element $f(x) \in B$. Both words matter: *every* input receives an output, and it receives *exactly one*. Different inputs may receive the same output, and some elements of B may never be used.

Think of an arrow leaving each input. Zero outgoing arrows or two different outgoing arrows from one input would violate the definition.

2.1 Domain, Codomain, Range, Image, and Preimage

Consider $A = \{a, b, c, d\}$, $B = \{1, 2, 3, 4\}$, and

$$f(a) = 2, \quad f(b) = 1, \quad f(c) = 2, \quad f(d) = 3.$$



- The **domain** is A , the set of allowed inputs.
- The **codomain** is B , the declared set of possible outputs.
- The **range**, or **image of the function**, is $f(A) = \{f(x) : x \in A\}$. Here it is $\{1, 2, 3\}$.

In these notes, “range” always means the outputs actually attained; the range is a subset of the codomain.

For a subset $U \subseteq A$, its **image** is

$$f(U) = \{f(x) : x \in U\}.$$

For a subset $V \subseteq B$, its **preimage** is

$$f^{-1}(V) = \{x \in A : f(x) \in V\}.$$

Thus $f(\{a, c\}) = \{2\}$, $f^{-1}(\{2\}) = \{a, c\}$, and $f^{-1}(\{2, 4\}) = \{a, c\}$. The preimage of $\{4\}$ is empty. An image goes forward along arrows; a preimage collects all inputs whose arrows land in the specified set.

Notation warning. The preimage $f^{-1}(V)$ is defined for *every* function. It is a set, and it does not require an inverse function. We will use $f^{-1}(y)$ for a single value only when an inverse function exists; $f^{-1}(\{y\})$ is the entire preimage set.

2.2 One-to-One, Onto, and Bijective

A function $f : A \rightarrow B$ is **one-to-one**, or **injective**, if different inputs have different outputs. Equivalently,

$$\forall x, x' \in A, \quad f(x) = f(x') \implies x = x'.$$

Each element of B has *at most one* preimage. To prove injectivity, start with equal outputs and show that the inputs must be equal. To disprove it, find two different inputs with the same output.

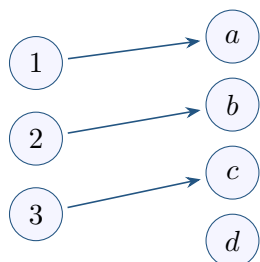
The function is **onto**, or **surjective**, if every element of its codomain is attained:

$$\forall y \in B \exists x \in A \quad f(x) = y.$$

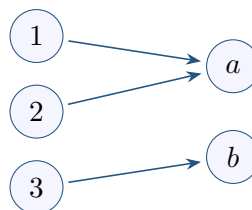
Thus $f(A) = B$, and each element of B has *at least one* preimage. To prove this, take an arbitrary target $y \in B$ and find a valid input. To disprove it, exhibit one target that is missed.

A **bijection** is both injective and surjective. Every element of B then has *exactly one* preimage: the function pairs the two sets without collisions or unused targets.

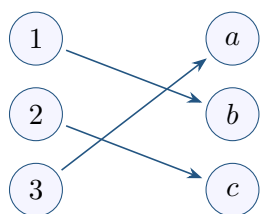
One-to-one, not onto



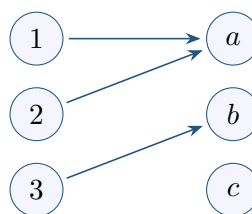
Onto, not one-to-one



Bijjective



Neither



Finite sizes. If A and B are finite, an injection requires $|A| \leq |B|$, a surjection requires $|A| \geq |B|$, and a bijection requires $|A| = |B|$. If the sets have equal finite size, an injection is automatically onto: distinct outputs already use every target. Likewise, an onto function cannot have a collision without missing another target.

Example: The Squaring Function

Consider the rule $f(x) = x^2$. We will use it to illustrate both one-to-one and onto functions.

One-to-one on the natural numbers. Let $f : \mathbb{N} \rightarrow \mathbb{N}$, where $\mathbb{N} = \{0, 1, 2, \dots\}$. For example,

$$0 \mapsto 0, \quad 1 \mapsto 1, \quad 2 \mapsto 4, \quad 3 \mapsto 9.$$

This function is one-to-one. If $f(x) = f(y)$, then $x^2 = y^2$. Since x and y are nonnegative, taking square roots gives $x = y$. It is not onto $\mathbb{N}$, since 2 is not a perfect square.

Onto the nonnegative real numbers. Now let $f : \mathbb{R} \rightarrow \mathbb{R}_+$, where $\mathbb{R}_+ = [0, \infty)$ includes zero. Given any $y \in \mathbb{R}_+$, choose the real number $x = \sqrt{y}$. Then

$$f(x) = (\sqrt{y})^2 = y.$$

Every element of the codomain is reached, so this function is onto. For example, the target 2 is now reached by the input $\sqrt{2}$. However, it is not one-to-one: $f(-2) = f(2) = 4$.

The rule is the same in both examples. Changing the domain and codomain changes whether the function is one-to-one or onto.

3 Composition and Inverses

3.1 Composition: Following One Function by Another

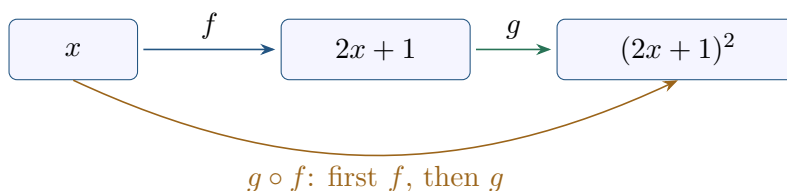
Given $f : A \rightarrow B$ and $g : B \rightarrow C$, their **composition** $g \circ f : A \rightarrow C$ is defined by

$$(g \circ f)(x) = g(f(x)).$$

Read from right to left: apply f first, then g . More generally, to define $g \circ f$ on all of A , every output of f must be an allowed input for g : $f(A) \subseteq \text{dom}(g)$.

Example 1: The Order Changes the Answer

Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$ be $f(x) = 2x + 1$ and $g(x) = x^2$.



$$(g \circ f)(x) = (2x + 1)^2 = 4x^2 + 4x + 1,$$

$$(f \circ g)(x) = 2x^2 + 1.$$

At $x = 2$, the answers are 25 and 9. Thus $g \circ f \neq f \circ g$: composition need not be commutative, even when both orders are defined.

Example 2: A Shift and a Reciprocal

Let $s : \mathbb{R} \rightarrow \mathbb{R}$ be $s(x) = x + 1$ and let $r : \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R} \setminus \{0\}$ be $r(x) = 1/x$. Following the two orders gives

$$r(s(x)) = \frac{1}{x+1} \quad (x \neq -1), \quad s(r(x)) = \frac{1}{x} + 1 \quad (x \neq 0).$$

The expressions have different values and different domains. Strictly, $r \circ s$ is not defined on all of the stated domain $\mathbb{R}$ of s , because $s(-1) = 0$. Restricting the input set to $\mathbb{R} \setminus \{-1\}$ makes that composition a function. In the other order, every output of r is a valid input for s .

Example 3: Square Roots and Domain Restrictions

Let $v : [0, \infty) \rightarrow [0, \infty)$ be $v(t) = \sqrt{t}$ and let $u : \mathbb{R} \rightarrow \mathbb{R}$ be $u(x) = x^2 - 4$. The expression

$$v(u(x)) = \sqrt{x^2 - 4}$$

is real exactly when $x^2 - 4 \geq 0$, that is, $x \in (-\infty, -2] \cup [2, \infty)$. This restriction is needed before we can treat $v \circ u$ as a function. In the other order,

$$(u \circ v)(x) = (\sqrt{x})^2 - 4 = x - 4 \quad (x \geq 0).$$

Although the simplified expression $x - 4$ makes sense for negative x , the composition still has domain $[0, \infty)$.

How to check a composition. First check that x is in the domain of the inner function. Then check that its output is in the domain of the outer function. Simplify only after recording these restrictions.

Associativity. For compatible functions, $h \circ (g \circ f) = (h \circ g) \circ f$, since both sides send x to $h(g(f(x)))$. Parentheses can change; the order cannot in general.

3.2 Inverses Defined by Composition

The **identity function** on A is $\text{id}_A : A \rightarrow A$, with $\text{id}_A(x) = x$. A function $h : B \rightarrow A$ is an **inverse** of $f : A \rightarrow B$ when

$$\boxed{h \circ f = \text{id}_A \quad \text{and} \quad f \circ h = \text{id}_B.}$$

Both compositions must give back their starting input. We write $h = f^{-1}$ when this happens.

Claim 1. *A function has a two-sided inverse if and only if it is a bijection.*

Proof. Suppose h is an inverse. If $f(x) = f(x')$, applying h gives $x = h(f(x)) = h(f(x')) = x'$, so f is injective. For any $y \in B$, $f(h(y)) = y$, so f is onto.

Conversely, if f is a bijection, each $y \in B$ has a unique preimage. Define $h(y)$ to be that preimage. Then $f(h(y)) = y$ for every $y \in B$, and $h(f(x)) = x$ for every $x \in A$. These are the two required identities. $\square$

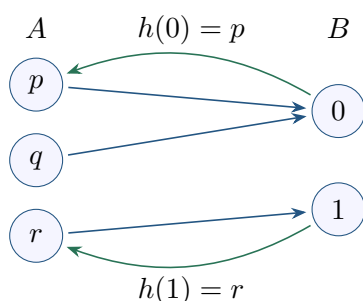
Checking both directions. For $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = 2x + 1$, the inverse is $h(y) = (y - 1)/2$, since

$$h(f(x)) = \frac{(2x + 1) - 1}{2} = x, \quad f(h(y)) = 2\frac{y - 1}{2} + 1 = y.$$

For the bijection $q : [0, \infty) \rightarrow [0, \infty)$, $q(x) = x^2$, the inverse is $q^{-1}(y) = \sqrt{y}$. The domain restriction matters: $\sqrt{x^2} = |x|$, which equals x on $[0, \infty)$ but not on all of $\mathbb{R}$.

An Onto Function May Only Be Undone in One Direction

If f is onto but not injective, a target can have several preimages. Choosing one for each target can give a **right inverse** $h : B \rightarrow A$, satisfying $f \circ h = \text{id}_B$, without satisfying the other identity. For example, let $f : \{p, q, r\} \rightarrow \{0, 1\}$ have $f(p) = f(q) = 0$ and $f(r) = 1$.



Blue: $f : A \rightarrow B$

Green: one choice of $h : B \rightarrow A$

$f(h(y)) = y$ for every $y \in B$

With $h(0) = p$ and $h(1) = r$, we have $f(h(y)) = y$ for both targets. But $h(f(q)) = p \neq q$. No function can recover both p and q from the same output 0. Thus a right inverse need not be a two-sided inverse.

3.3 Composing Bijections

If $f : A \rightarrow B$ and $g : B \rightarrow C$ are bijections, then $g \circ f : A \rightarrow C$ is also a bijection. To see injectivity, suppose $g(f(x)) = g(f(x'))$. Injectivity of g gives $f(x) = f(x')$; injectivity of f gives $x = x'$. To see surjectivity, take $z \in C$. Surjectivity of g gives $y \in B$ with $g(y) = z$, and surjectivity of f gives $x \in A$ with $f(x) = y$. Thus $g(f(x)) = z$.

Its inverse is

$$(g \circ f)^{-1} = f^{-1} \circ g^{-1}.$$

To undo “first f , then g ,” undo g first and then f . This lets us compare sets by passing through an intermediate set.

4 Countably Infinite Sets

Throughout this lecture,

$$\mathbb{N} = \{0, 1, 2, \dots\}, \quad \mathbb{N}_{>0} = \{1, 2, 3, \dots\}.$$

Two sets have the **same cardinality** if there is a bijection between them. An infinite set S is **countably infinite** if there is a bijection $e : \mathbb{N} \rightarrow S$. Such a function lists its elements as $e(0), e(1), e(2), \dots$, with every element appearing exactly once. A set is called **countable** if it is finite or countably infinite.

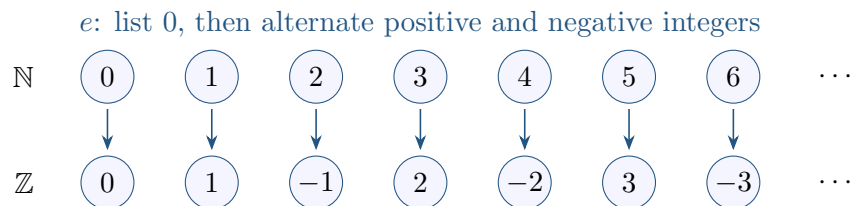
To prove countable infinitude, construct a bijection with $\mathbb{N}$ and check both parts: no repeated outputs (injectivity) and no missing elements (surjectivity). A bijection in the reverse direction works too, because its inverse is a bijection.

4.1 Revisiting the Pairing of Natural Numbers and Integers

Earlier, we paired the integers with natural numbers by alternating positive and negative values. We can now express that pairing as a function $e : \mathbb{N} \rightarrow \mathbb{Z}$:

$$e(n) = \begin{cases} -n/2, & n \text{ is even,} \\ (n+1)/2, & n \text{ is odd.} \end{cases}$$

For even $n = 2k$, the output is $-k$; for odd $n = 2k + 1$, it is $k + 1$.



Claim 2. *The function $e : \mathbb{N} \rightarrow \mathbb{Z}$ is a bijection.*

Proof. One-to-one. Suppose $e(m) = e(n)$. An even input has a nonpositive output, while an odd input has a positive output, so m and n must have the same parity. If they are even, the equality $-m/2 = -n/2$ gives $m = n$. If they are odd, the equality $(m+1)/2 = (n+1)/2$ again gives $m = n$.

Onto. Take any $z \in \mathbb{Z}$. If $z \leq 0$, choose the even natural number $n = -2z$; then $e(n) = z$. If $z > 0$, choose the odd natural number $n = 2z - 1$; then $e(n) = z$. Every integer is attained. $\square$

The proof also gives the inverse:

$$e^{-1}(z) = \begin{cases} -2z, & z \leq 0, \\ 2z - 1, & z > 0. \end{cases}$$

For example, $e(7) = 4$, $e(8) = -4$, and $e^{-1}(-6) = 12$. Thus $|\mathbb{Z}| = |\mathbb{N}|$: adding all the negative integers does not produce a larger cardinality.

4.2 Encoding a Pair of Natural Numbers as One Positive Integer

The set $\mathbb{N}^2 = \mathbb{N} \times \mathbb{N}$ consists of all ordered pairs (a, b) of natural numbers. It looks like an infinite grid, with infinitely many rows and infinitely many columns. Can every point receive its own natural-number label, with every label used exactly once?

We first use *positive* labels. Define $F : \mathbb{N}^2 \rightarrow \mathbb{N}_{>0}$ by

$$F(a, b) = 2^a(2b + 1).$$

The first coordinate chooses a power of two; the second chooses an odd positive integer. Their product is always a positive integer, so the rule is well-defined on all of $\mathbb{N}^2$.

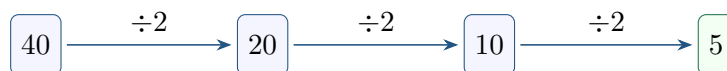
		b						
		0	1	2	3	4	5	
	0	1	3	5	7	9	11	$\cdots \quad 2^0 \times \text{odd}$
	1	2	6	10	14	18	22	$\cdots \quad 2^1 \times \text{odd}$
a	2	4	12	20	28	36	44	$\cdots \quad 2^2 \times \text{odd}$
	3	8	24	40	56	72	88	$\cdots \quad 2^3 \times \text{odd}$
	4	16	48	80	112	144	176	$\cdots \quad 2^4 \times \text{odd}$
		$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	
		Entry in row a , column b : $F(a, b) = 2^a(2b + 1)$						

Row $a = 0$ contains all odd positive integers. Row $a = 1$ contains numbers divisible by 2 but not 4; row $a = 2$ contains numbers divisible by 4 but not 8. In general, row a contains numbers whose factorization has *exactly* a factors of 2.

Encode and decode. For example,

$$F(3, 2) = 2^3(2 \cdot 2 + 1) = 40.$$

To recover the pair from 40, divide by 2 until an odd number remains:



Three divisions give $a = 3$; the odd factor $5 = 2b + 1$ gives $b = 2$.

There were three divisions, so $a = 3$. The remaining odd number is $5 = 2b + 1$, so $b = 2$. The two coordinates are stored in different parts of the factorization: the power of 2 and the odd factor.

Claim 3. *The map $F : \mathbb{N}^2 \rightarrow \mathbb{N}_{>0}$, $F(a, b) = 2^a(2b + 1)$, is one-to-one and onto.*

Proof. One-to-one. Suppose two pairs have the same image:

$$2^a(2b + 1) = 2^c(2d + 1).$$

If $a < c$, divide both sides by 2^a to obtain

$$2b + 1 = 2^{c-a}(2d + 1).$$

The left side is odd, while the right side is even because $c - a \geq 1$. This is impossible. The same argument rules out $c < a$. Therefore $a = c$. Canceling 2^a now gives $2b + 1 = 2d + 1$, hence $b = d$. Thus the original pairs are equal, proving injectivity.

Onto (strong induction). We induct on the positive integer m .

Base case: $m = 1$. We have $F(0, 0) = 2^0(2 \cdot 0 + 1) = 1$.

Inductive step: Let $m > 1$ and assume every positive integer smaller than m has a preimage.

- If m is odd, choose $a = 0$ and $b = (m - 1)/2 \in \mathbb{N}$. Then $F(a, b) = 2^0(2b + 1) = m$.
- If m is even, then $1 \leq m/2 < m$. By the induction hypothesis, $m/2 = F(a, b) = 2^a(2b + 1)$ for some $a, b \in \mathbb{N}$. Hence

$$F(a + 1, b) = 2^{a+1}(2b + 1) = 2 \cdot \frac{m}{2} = m.$$

In either case, m has a preimage. By strong induction, F is onto. □

Thus each positive integer has a unique representation $2^a(2b + 1)$ with $a, b \in \mathbb{N}$.

There is a simple bijection $h : \mathbb{N}_{>0} \rightarrow \mathbb{N}$:

$$h(m) = m - 1, \quad h^{-1}(n) = n + 1.$$

Indeed, $h(m) = h(m')$ gives $m - 1 = m' - 1$, hence $m = m'$, so it is injective. Given any $n \in \mathbb{N}$, the positive integer $m = n + 1$ satisfies $h(m) = n$, so it is onto. Notice the endpoint: $h(1) = 0$.

Composing the two bijections gives

$$\mathbb{N}^2 \xrightarrow{F} \mathbb{N}_{>0} \xrightarrow{h} \mathbb{N}, \quad H(a, b) = 2^a(2b + 1) - 1.$$

By our result about compositions of bijections, H is a bijection. Therefore $|\mathbb{N}^2| = |\mathbb{N}|$, and $\mathbb{N}^2$ is countably infinite.